INDIAN INSTITUTE OF TECHNOLOGY JODHPUR | B.TECH PROJECT EVALUATION (2026)

Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates

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Background & Engineering Motivation

The Construction & Demolition (C&D) Waste Crisis

- **Global Aggregate Demand:** Over 50 billion tonnes/year of virgin stone & sand mined globally, depleting quarries and riverbeds.
- **C&D Waste Generation:** Exceeds 3.0 billion tonnes/year worldwide; India generates >150 million tonnes/year (CPCB 2024), yet <5% is reused structurally.
- **Circular Economy Materials:** Recycled Concrete (RC), Recycled Mortar (RM), and Recycled Brick (RB) offer sustainable substitutes for highway subgrades & concrete.

Why Recycled Aggregates Crush Prematurely

- **Multi-Phase Heterogeneity:** Unlike virgin basalt/granite, recycled aggregate retains porous adhered mortar & two weak Interfacial Transition Zones (ITZs).
- **Flaw & Water Vulnerability:** High micro-crack density gives 4–10% water absorption (vs. <1.5% virgin rock) and wide strength scatter.
- **Compaction Degradation:** Under vibratory rollers & traffic loads, grains crush into excess fines (<0.075 mm), causing drainage blockages & rutting.

Core Goal: Screen and predict single-particle crushing mechanics & post-breakage gradation non-destructively in <0.1 seconds before road/concrete placement.

Problem Definition & Research Objectives

Evaluation Aspect	Traditional Lab Testing (UTM + Sieving / IS 2386)	Proposed Physics-Informed ML Suite
Test Nature	100% Destructive — specimen lost after single crush	Non-Destructive — zero sample destruction
Testing Speed	16–24 hours per batch (calipers, UTM, mechanical sieving)	< 0.1 seconds per particle (real-time inference)
Occupational Safety	Generates hazardous respirable crystalline silica dust	Completely digital & dust-free screening
Shape Sensitivity	Coarse bulk indices (ACV/LAA) hide 3D shape effects	Captures continuous 3D Sphericity (Φ) & L, I, S axes
Output Completeness	Single aggregate strength or isolated sieve value	Simultaneous Force, Strength, Energy & Fractal Curve

Four Core Project Objectives (Within 4.75–37.5 mm Quasi-Static Scope)

- **1. Physics-Coupled Benchmark Dataset:** Synthesize N = 3,600 calibrated particles (RC, RM, RB) uniting Hiramatsu-Oka, Bagheri, Cai et al. (2024), and Liang et al. (2025) fracture laws.
- **2. Multi-Target ML Benchmarking:** Compare 6 regressors (CatBoost, LightGBM, XGBoost, Random Forest, MLP Neural Network, Ridge) across Force, Strength, Energy & Fractal Dimension.
- **3. Explainable AI (TreeSHAP) Verification:** Quantify physical feature importance on XGBoost models to confirm they obey brittle fracture physics rather than overfitting noise.
- **4. Interactive Streamlit Engineering Suite:** Deploy a 4-tab tool (Virtual UTM, Scaling Simulator, Auxiliary CV Photo Scanner, AI & Benchmarks) for non-destructive aggregate screening.

Governing Fracture Physics & 3D Morphology

Geometry & Tensile Rupture (Bagheri & Hiramatsu-Oka)

- **Triaxial Dimensions:** Longest (L), Intermediate (I), Shortest (S) axes; 3D Sphericity $\Phi = \sqrt[3]{[(I \cdot S)/L^2]}$
- **Effective Diameter (Eq. 4.2):** $d_o = 0.887 \cdot \sqrt[3]{(L \cdot I \cdot S)}$ [Bagheri et al., 2015]
- **Peak Force–Strength (Eq. 4.3):** $Ff = (\sigma f \cdot d_o^2) / 0.9$ [Hiramatsu & Oka, 1966]

Weibull Scaling & Fractal Gradation (Liang / Cai / Turcotte)

- **Strength Scaling (Eq. 4.4):** $\sigma f(d_o, \Phi) = [p \cdot \Phi^q] \times [\sigma f_o(d_o) / \sigma f_o(d_ref)]$ (Liang Table 2)
- **Crushing Energy (Eq. 4.5):** $Ef = c \cdot \Phi^d \cdot (d_o/25)^3$ ($\Phi<0.8$) | Sat. Law ($\Phi \geq 0.8$) [mJ]
- **Fractal Gradation (Eq. 4.6):** $F(d_i) = \min[100, (d_i / d_o)^{(3 - D)} \times 100]$

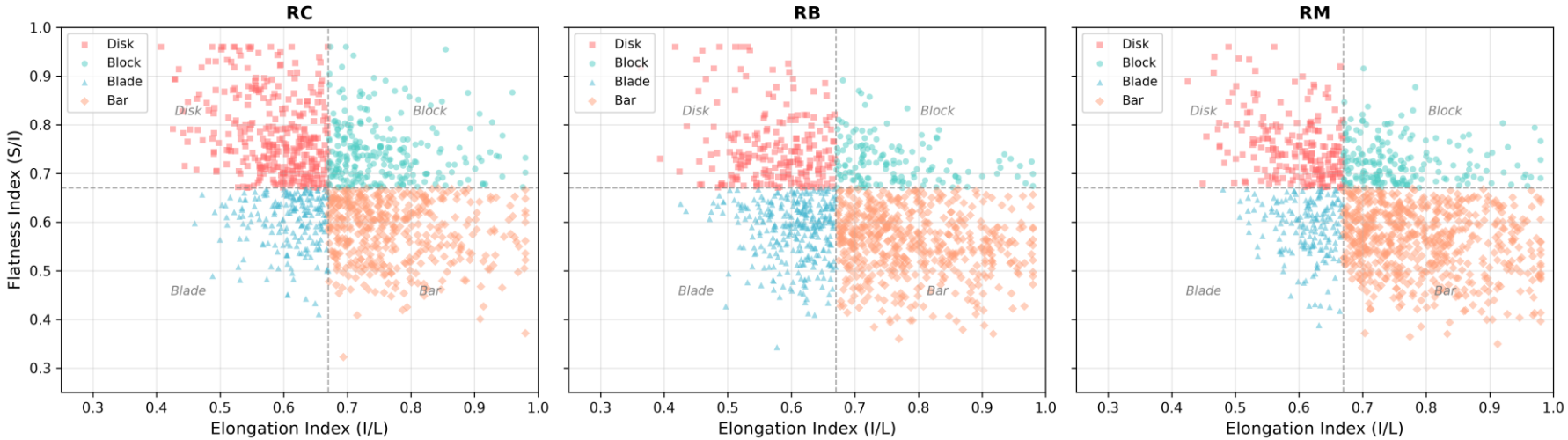


Fig. 4.1: Zingg 3D particle shape classification across Disk, Block, Blade, and Bar morphological domains (0.67 cutoffs; N=3,600).

Physics-Coupled Calibration Dataset (N = 3,600)

Material Type	σf_0 (MPa) [3 brk]	m [3 brackets]	p , q	c (mJ), d	D_size [3 brk]
Recycled Concrete (RC)	6.51 / 6.28 / 6.07	1.58 / 1.30 / 1.25	9.35, 2.13	2061, 1.28	1.66–1.87
Recycled Mortar (RM)	3.03 / 2.97 / 2.91	2.70 / 2.73 / 2.76	4.69, 2.21	2041, 2.66	1.77–2.11
Recycled Brick (RB)	6.35 / 6.21 / 4.92	2.58 / 2.61 / 3.32	3.84, 1.68	1511, 2.16	1.68–2.01

Benchmark Dataset Architecture (Table 4.1)

- **Total Samples:** N = 3,600 synthetic particles calibrated strictly to Liang et al. (2025) & Cai et al. (2024) tables.
- **Three Sieve Brackets:** 400 particles per bracket × 3 materials across 4.75–9.5 mm, 9.5–19.0 mm, and 19.0–37.5 mm.
- **Material Ranking:** RC has highest shape strength (p = 9.35 MPa, dataset mean 4.30 MPa); porous RB (mean 2.31 MPa) & RM (2.11 MPa) are weaker.
- **Experimental Noise:** Realistic ~7% measurement noise injected across targets matching laboratory UTM dispersion.

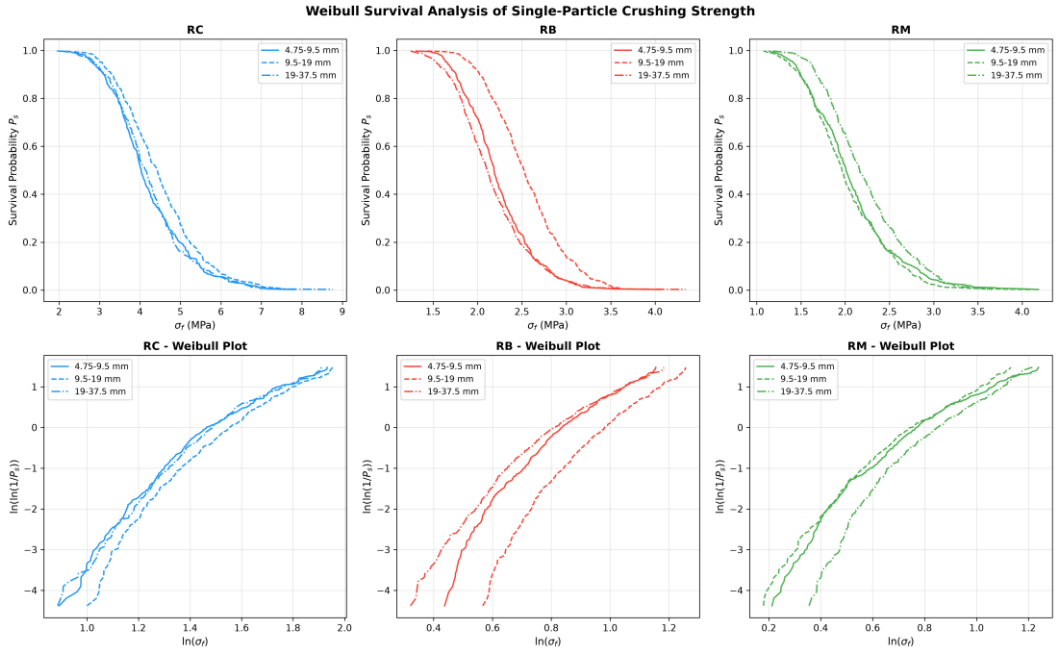


Fig. 4.2: Weibull survival distributions and log-linear plots across 3 materials and sieve brackets.

Machine Learning Architecture & Training Pipeline

1. 16-D Feature Vector

13 Continuous Inputs:
L, I, S, EI, FI, Φ, Sf, d_eq, d_o,
App./Bulk Density, E, ν
+ 3 Label Encoded Classes

2. Stratified Partitioning

80% Training (N = 2,880)
20% Hold-out Test (N = 720)
StandardScaler Z-Score +
Breakage-Mode Stratification

3. Six Trained Regressors

CatBoost, LightGBM,
XGBoost, Random Forest,
MLP Neural Network,
& Ridge Polynomial (deg=2)

4. Four Target Outputs

Crushing Strength σf (MPa)
Peak Force Ff (kN)
Crushing Energy Ef (mJ)
Fractal Dimension D

Model Architecture	Key Hyperparameters	Loss Function	Regularization	Optimization
CatBoost Regressor	depth=8, iterations=300, lr=0.05	**RMSE**	**l2_leaf_reg=3.0**	**Ordered Boosting**
LightGBM Regressor	max_depth=8, n_est=300, lr=0.05	Squared Error	reg_alpha=0.1, λ=1.0	Histogram GBDT
XGBoost Regressor	max_depth=8, n_est=300, lr=0.05	Squared Error	colsample=0.8, sub=0.8	Hist / Greedy Tree
Random Forest	n_estimators=200, max_depth=15	Squared Error	min_split=5, min_leaf=2	Bagging Ensemble
MLP Neural Network	hidden=(128,64,32), ReLU	MSE (Early Stop)	alpha=0.001, batch=64	Adam (lr=1e-3)
Ridge Polynomial	degree=2, include_bias=False	Squared Loss	alpha=1.0 (L2 Penalty)	Closed-Form Ridge

Model Performance Benchmarks on Hold-Out Test Set

Model Architecture	Target Variable	R^2 Score	RMSE	MAE	MAPE (%)
CatBoost Regressor	Peak Force Ff (kN)	**0.9871**	**0.081 kN**	**0.050 kN**	**6.56%**
CatBoost Regressor	Crushing Strength (MPa)	0.9636	0.235 MPa	0.172 MPa	5.81%
CatBoost Regressor	Crushing Energy Ef (mJ)	0.9498	82.53 mJ	23.36 mJ	9.13%
CatBoost Regressor	Fractal Dimension D	0.9469	0.032	0.025	1.37%
LightGBM Regressor	Peak Force Ff (kN)	0.9852	0.087 kN	0.052 kN	6.48%
XGBoost Regressor	Peak Force Ff (kN)	0.9847	0.089 kN	0.054 kN	6.67%
Random Forest	Peak Force Ff (kN)	0.9840	0.091 kN	0.054 kN	6.80%
MLP Neural Network	Peak Force Ff (kN)	0.9866	0.083 kN	0.052 kN	7.66%
Ridge Polynomial	Energy Ef (Piecewise Fail)	0.8647	135.44 mJ	54.40 mJ	65.08%

- **Why CatBoost Leads Overall:** CatBoost achieves top mean R^2 = 0.9619 (0.9871 Force, 0.9636 Strength). Ridge collapses on piecewise energy Ef (MAPE 65.1%), proving GBDT necessity.

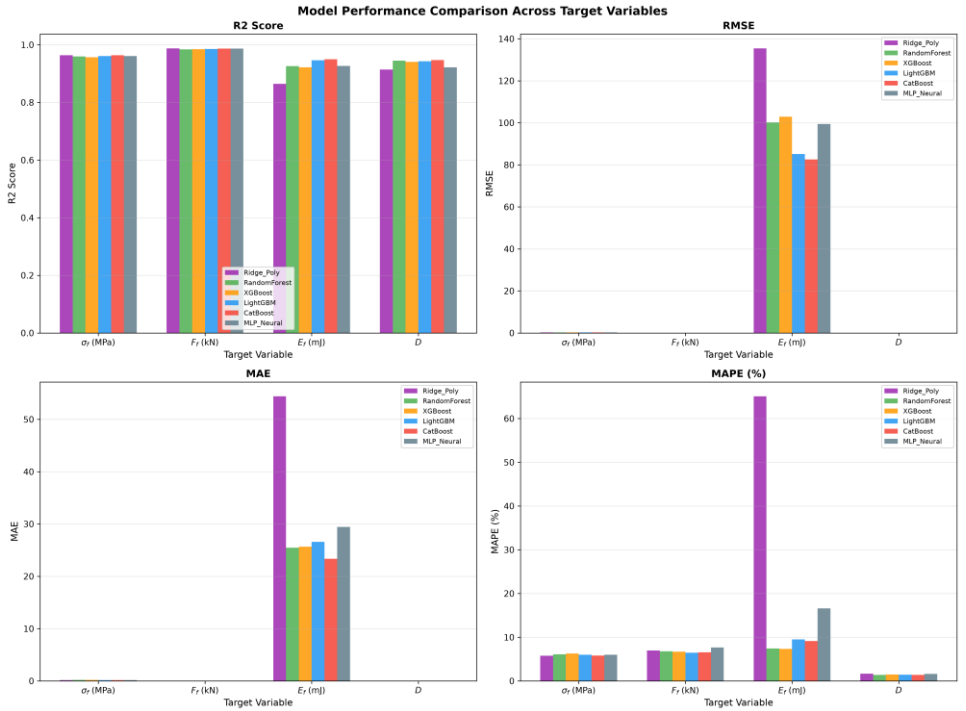


Fig. 5.0: Comparative R^2, RMSE, MAE, and MAPE across the 6 regression architectures.

Actual vs. Predicted Parity & Residual Error Analysis

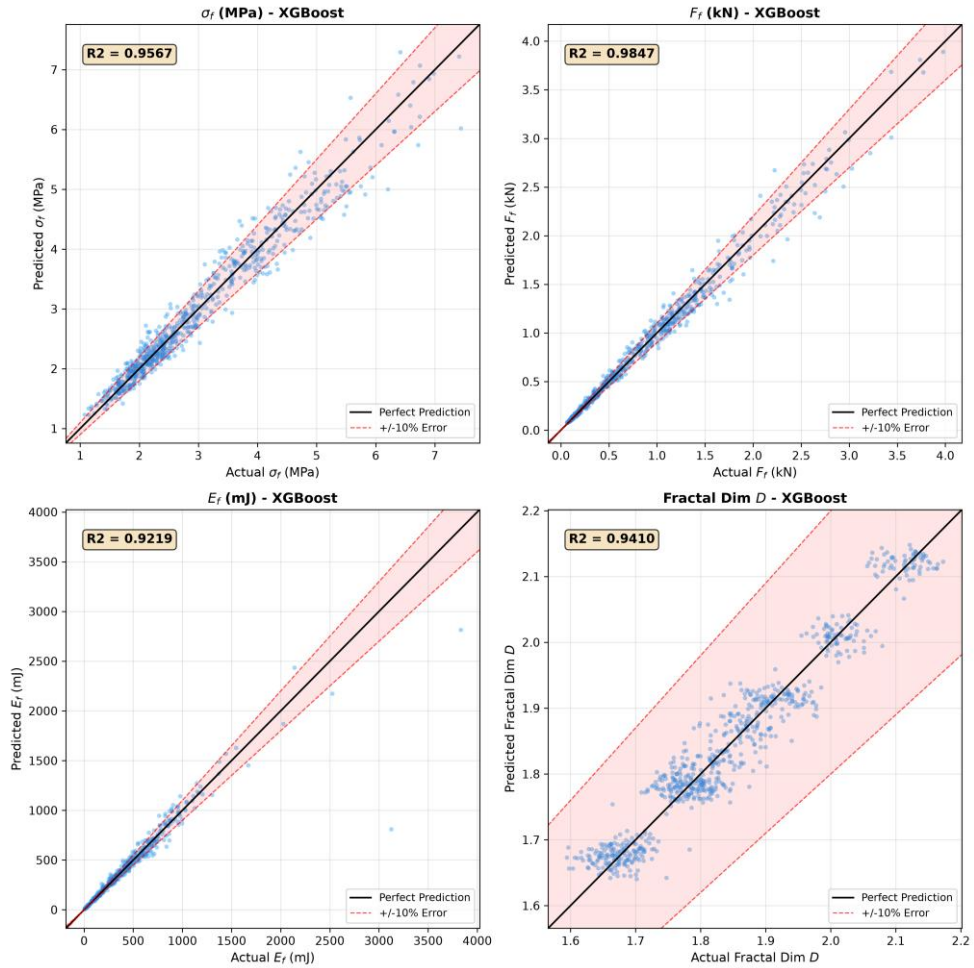


Fig. 5.1: XGBoost test-set parity plots across σf, Ff, Ef, and D (±10% band).

Parity Diagnostics (Fig. 5.1 — XGBoost R^2 = 0.922–0.985)

- **Tight ±10% Parity:** Most test points stay within the ±10% error band across all 3 materials (R^2 = 0.957 σf, 0.985 Ff, 0.922 Ef, 0.941 D).
- **Clustered Fractal Tiers:** Fractal D plot reflects the 3 calibrated Liang et al. (2025) sieve brackets around 1.66–2.12.

CatBoost Residual Stability across Sieve Brackets (Table 5.2)

Size Bracket	σf Mean / Std	Ff Mean / Std	Variance Check
Fine (4.75–9.5 mm)	-0.013 / 0.230	-0.001 / 0.018	σf Homoscedastic
Medium (9.5–19 mm)	+0.002 / 0.245	+0.001 / 0.067	σf Levene p = 0.924
Coarse (19–37.5 mm)	+0.043 / 0.228	+0.020 / 0.120	Unbiased Mean ~0

Note: Crushing strength σf is strictly homoscedastic (p = 0.924); Ff std scales with d_o^2 while relative MAPE stays ~6.6%.

Explainable AI: Physical Attribution via TreeSHAP

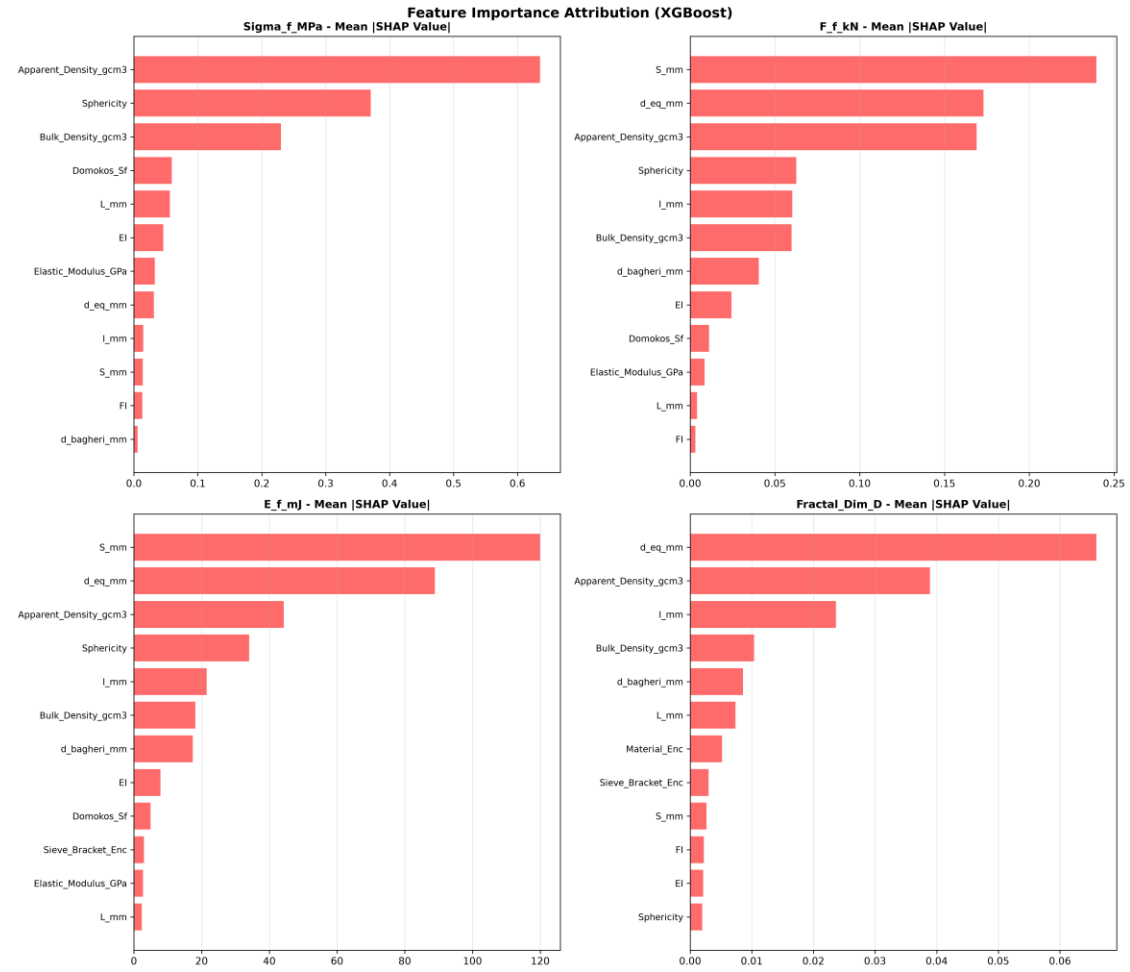


Fig. 5.2: XGBoost TreeSHAP Mean |SHAP| feature importance across all 4 targets.

Deciphering the Physics Inside XGBoost (TreeSHAP)

- **1. Strength σf — Density & Sphericity Φ:** Apparent density (#1) separates strong RC (2.22 g/cm³) from RB/RM, while 3D Sphericity Φ (#2) captures platen stress distribution.
- **2. Force Ff & Energy Ef — Size Axes (S, d_eq):** Shortest axis S (#1) and d_eq (#2) dominate Ff and Ef directly matching $Ff \propto d_o^2$ area scaling and $(d_o/25)^3$ volume scaling.
- **3. Fractal Dimension D — Size & Material:** d_eq (#1) and Apparent Density (#2) drive post-breakage D because larger fragments generate more secondary debris.
- **4. No Unphysical Shortcut Learning:** TreeSHAP confirms the ML splits strictly follow Hiramatsu-Oka, Cai, and Liang fracture mechanics.

2D Partial Dependence & Coupled Scaling Surfaces

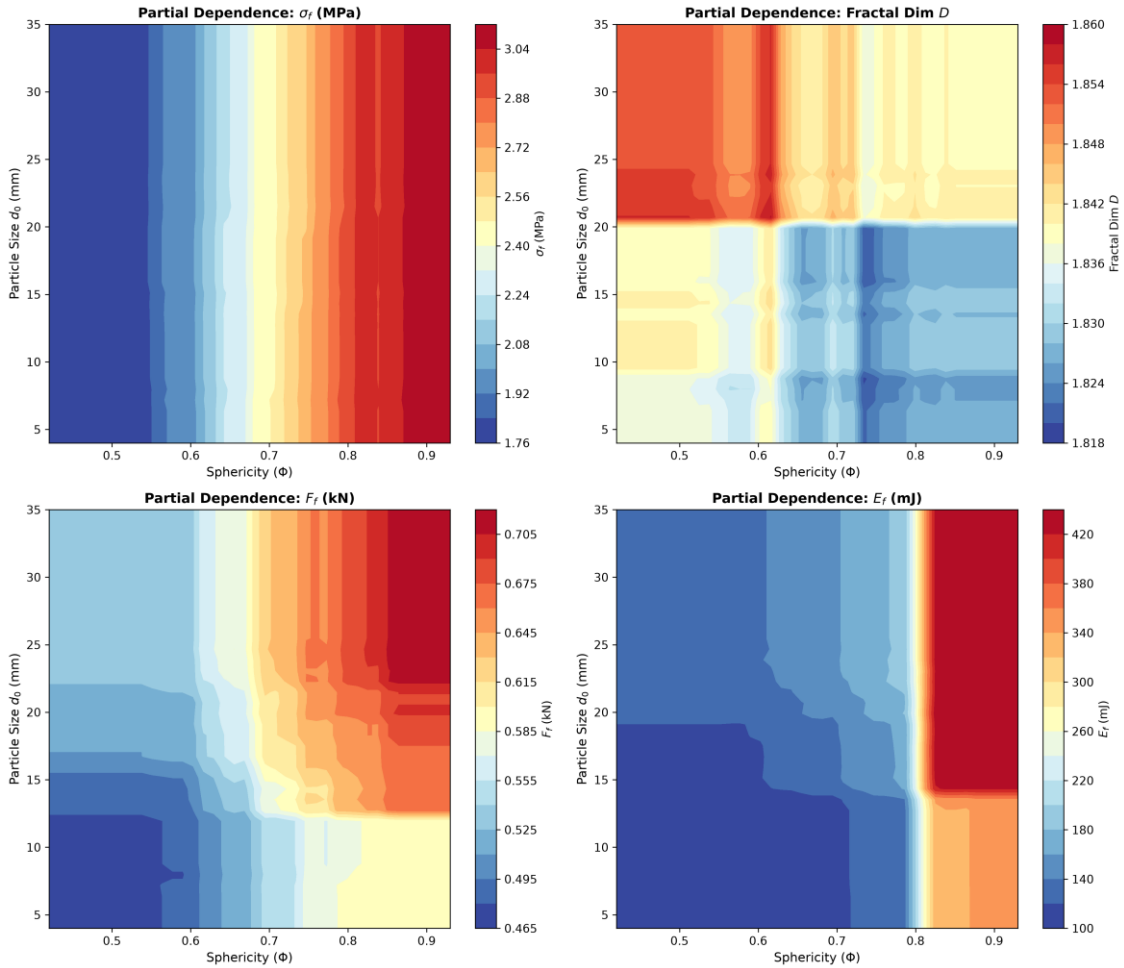


Fig. 5.3: 2D Partial Dependence contour heatmaps over diameter d_o and sphericity Φ.

Engineering Design Insights from Contours

- **Coupled (Φ, d_o) Contours:** Strength σf rises steadily with Sphericity Φ (left to right), while Peak Force Ff peaks at upper-right where both size d_o and Φ are high.
- **Piecewise Transition at Φ = 0.80:** Crushing energy Ef plot exposes the sharp Cai et al. (2024 Eq. 11) jump at Φ = 0.80 learned directly by the ML trees.
- **Step Boundary at d_o = 20 mm:** Fractal D heatmap reveals the discrete sieve-bracket step at 20 mm in raw training bins—motivating smooth Tab 2 interpolation.
- **Quality Control Guideline:** Avoiding flaky particles (Φ < 0.55) prevents low strength and premature brittle failure.

Interactive Virtual Testing Suite (Streamlit Software Demo)

Tab 1: Virtual UTM & Sieve Gradation Curve

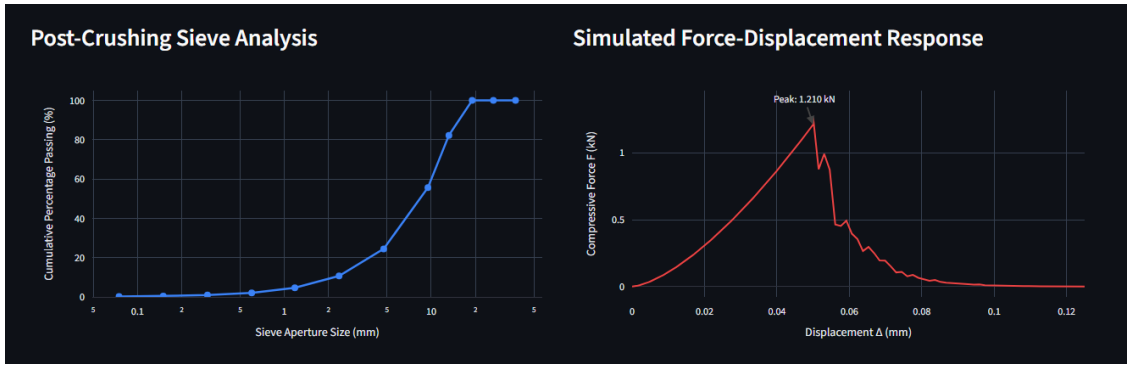


Fig. 6.1: Live single-particle virtual compression & ASTM/IS post-breakage PSD.

Tab 2: Multi-Scale Scaling Contour Simulator

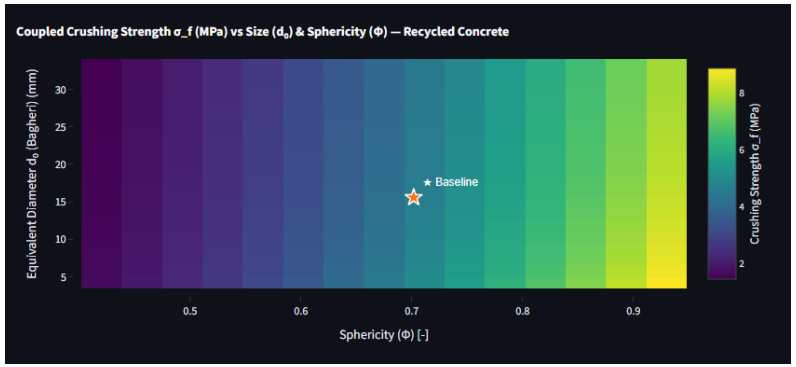
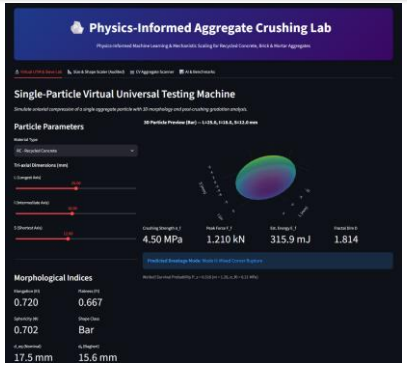


Fig. 6.2: Interactive 2D size-scaling response surfaces.

Parametric Console & 3D Preview (Tab 1)



Material selector, triaxial sliders (25 × 18 × 12 mm RC), and live 3D ellipsoid.

Four Integrated Engineering Modules

- **Tab 1 — Virtual UTM:** Instant σf, Ff, Ef, D, Weibull P_s survival metric, breakage mode, and sieve curve.
- **Tab 2 — Scaling Simulator:** Continuous vs. discrete size-scaling contours benchmarked against lab checkpoints.
- **Tab 3 — Auxiliary CV Scanner:** Optional OpenCV photo sizer estimating 2D L & I (assumes S = 0.65·I) as a convenience helper.
- **Tab 4 — AI & Benchmarks:** Interactive R^2 bar chart + test metrics tables + all 9 publication plots (SHAP, PDP, Parity).

Baseline RC Case Study & Pavement Gradation Screening

Output Parameter	Value	Unit	Physical Interpretation
Input Particle Size	25 / 18 / 12	**mm**	**Recycled Concrete (L / I / S axes)**
Equivalent Diameter d_eq	17.54	mm	Equal-volume nominal diameter
Bagheri Diameter d_o	15.56	mm	Roughness-corrected diameter
FI (S/I) | EI (I/L)	0.667 | 0.720	—	Zingg Bar/Block boundary (0.67 cutoff)
3D Sphericity (Φ)	0.702	—	High sphericity (> 0.65 favorable zone)
Crushing Strength (σf)	4.50	MPa	Hiramatsu-Oka tensile splitting strength
Peak Failure Force (Ff)	1.210	kN	Catastrophic rupture load
Est. Crushing Energy (Ef)	315.9	mJ	Cai piecewise volumetric energy (0.316 J)
Fractal Dimension (D)	1.814	—	Single-particle exponent (Mode II rupture)

Gradation & Pavement Screening (Li 2021 & IRC:37-2018)

- **GSB / WMM Screening:** IRC:37-2018 & IS 383:2016 require roadbase aggregates to resist crushing into fine dust under roller compaction.
- **Over-Pulverization Risk (D > 2.63):** Above Li et al. (2021) upper limit D = 2.631, crushing produces excess <0.075 mm fines, hurting permeability & causing rutting.
- **Baseline RC Verdict (D = 1.814):** Single-particle D = 1.814 (composite RC = 2.12, inside Li 1.887–2.631) indicates coarse splitting with minimal fine silt generation.
- **Failure & Survival Check:** Mode II Mixed Corner Rupture with Weibull survival probability P_s = 0.516 (m = 1.28, σ_f0 = 6.21 MPa).

Validation Against Published Experimental Benchmarks

Material Type	Published Experimental Source	MAE (MPa)	RMSE (MPa)	MAPE (%)	Agreement
Recycled Concrete (RC)	Liang et al. (2025) & Cai et al. (2024)	0.043	0.060	1.15%	Outstanding
Recycled Mortar (RM)	Liang et al. (2025) & Cai et al. (2024)	0.030	0.047	2.02%	Outstanding
Recycled Brick (RB)	Liang et al. (2025) & Cai et al. (2024)	0.148	0.180	5.07%	High Fidelity

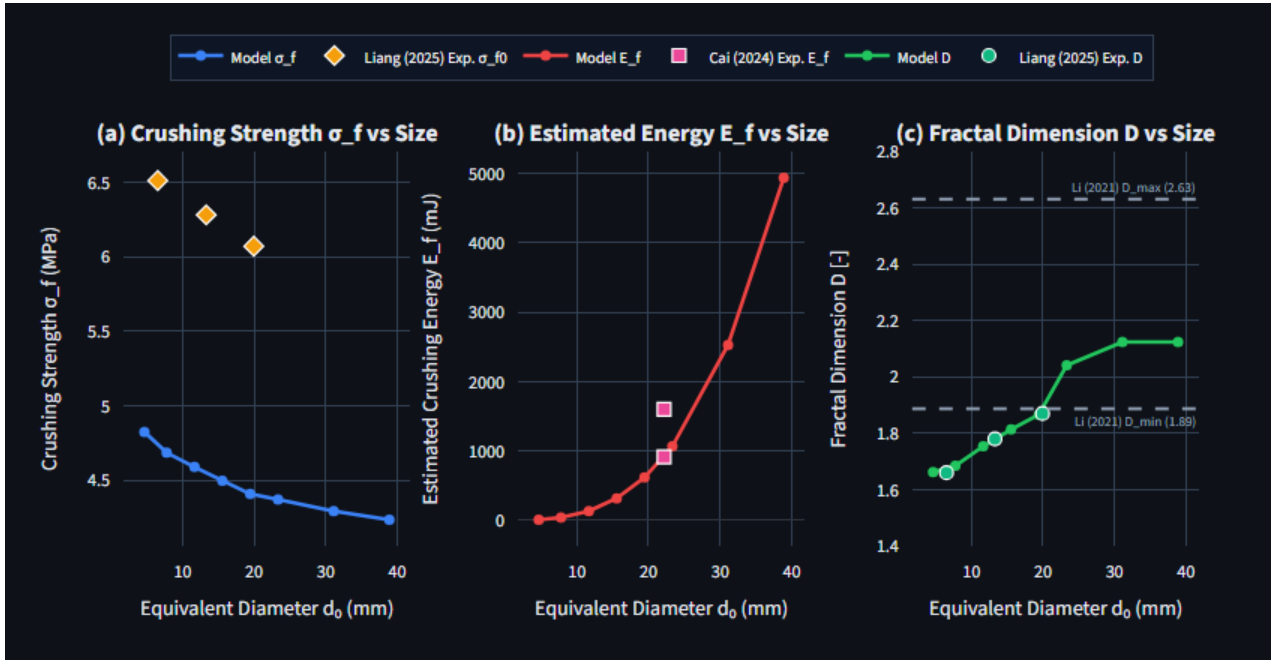


Fig. 6.3: Tab 2 size-scaling curves vs. experimental markers (Liang et al., 2025; Cai et al., 2024; Li et al., 2021).

Summary of Contributions & Future Work

Key Project Contributions

- **Unified PIML Dataset:** Coupled Weibull size scaling, Hiramatsu-Oka tensile laws, Cai 3D sphericity, and Turcotte fractal gradation into N = 3,600 benchmark particles.
- **High-Accuracy CatBoost Surrogate:** Achieved best average R^2 = 0.9619 (0.9871 Force, 0.9636 Strength, 0.9498 Energy, 0.9469 D) & homoscedastic σf residuals.
- **Transparent Explainable AI:** Verified physics via XGBoost TreeSHAP (Density & Φ govern strength σf; S & d_eq govern Force & Energy).
- **Deployable Engineering Suite:** Delivered a 4-tab Streamlit app (Tab 1 Virtual UTM: 1.210 kN / 4.50 MPa baseline) cutting test time to <0.1 s.

Known Limitations & Future Work

- **Current Boundaries:** Quasi-static single-particle compression (4.75–37.5 mm); calibrated on literature data; CV assumes S = 0.65·I; mode clf ~54% acc.
- **Indian C&D Waste Lab Testing:** Perform physical UTM crushing on locally processed demolition samples (Delhi, Ahmedabad, Jodhpur) to validate locally.
- **Bulk DEM Simulation Coupling:** Integrate ML breakage criteria into 3D DEM codes (PFC3D / YADE) for full roller compaction & cyclic traffic simulations.
- **3D Depth Photogrammetry:** Upgrade optional CV helper with multi-view depth sensing to measure shortest axis S directly without assumptions.

Thank You! | Open for Questions & Live Streamlit Virtual UTM Demonstration